

Christopher McGovern

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Summary

Senior Data Engineer with over 9 years of experience in the **telecom and software industries**, specializing in **data architecture, pipeline development, and large-scale data processing**. Expertise in designing and optimizing **scalable, high-performance data solutions**, leveraging technologies such as **Apache Spark, Hadoop, Kafka, AWS, GCP, Snowflake, DBT, and PostgreSQL**. Proficient in **ETL/ELT workflows, real-time and batch data processing, and cloud-native data engineering**. Skilled in leading **cross-functional teams**, collaborating with **stakeholders**, and driving **data-driven decision-making** by developing efficient, **secure, and reliable data infrastructures**. Passionate about building **scalable data ecosystems** to support **business intelligence, AI/ML models, and enterprise analytics**.

Skills

- **Big Data Technologies:** Apache Spark, Apache Hadoop, Apache Hive, Apache Flink, Apache Sqoop, Cloudera Data Platform, Hortonworks Data Platform, Apache Kylo, Apache Kafka, Apache Nifi, OpenSearch
- **Data Warehousing & ETL:** Snowflake, Amazon Redshift, PostgreSQL, Apache Kylin, DBT (Data Build Tool), Data Modeling, Google Big Query, Apache Iceberg
- **Cloud Services:** AWS (Redshift, S3, EC2, EMR, Athena, Glue, Lambda), GCP(Big Query, Dataflow, Pub/Sub)
- **Schedule & Workflow Monitoring:** Apache Airflow, Apache Oozie
- **Cluster Resource Management:** Apache Zookeeper, Apache YARN, Kubernetes
- **Machine Learning & NLP:** Regression, Classification, Clustering, Parts-of-Speech Tagging, Named Entity Recognition, Classifiers, Dependency Trees
- **Programming Languages:** Java (Core Java, J2EE, JDBC), Python, Scala, C#, C++, JavaScript, TypeScript
- **Web Technologies & Frameworks:** HTML, CSS, Angular 2, Entity Framework, Jersey Framework, GraphQL
- **Version Control:** GitKraken, GitHub, GitLab
- **Databases:** MySQL, SQL Server, Oracle 10g, PostgreSQL, MongoDB, Apache Cassandra
- **Operating Systems & DevOps:** Linux (Ubuntu, CentOS, Red Hat), Windows 8/10, Containerization(Docker, Kubernetes, Helm)
- **IDE & Tools:** Eclipse, PyCharm, MS SQL Server Management Studio, Visual Studio, NetBeans, Brackets, WebStorm, Dreamweaver
- **Web Servers:** Tomcat 7.0.x, WAMP Server

Experience

Atlassian

Senior Data Engineer

July 2023 – Present
San Francisco, CA

- Expert in designing and implementing large-scale **data architectures**, utilizing open-source technologies such as **Apache Spark, Apache Flink, Apache Kafka, Hadoop, Apache Iceberg, and Apache NiFi**. Skilled in both **SQL and NoSQL** databases, including **BigQuery, PostgreSQL, Cassandra, MongoDB, MySQL, and MSSQL**.
- Developed and optimized real-time **fraud detection systems** using **Apache Flink and Kafka**, applying advanced analytics, anomaly detection, and **AML-based matching criteria** to identify and mitigate fraudulent activities in high-volume transactional environments.
- Architected and managed large-scale **ETL pipelines** for real-time and batch data processing, automating data ingestion and transformation from multiple sources to centralized **data lakes**, improving **data accessibility** and efficiency.

- Led **cloud migration** strategies to **AWS**, optimizing **system scalability** and reducing **operational costs** by 20% through the adoption of **S3, Redshift, Lambda, and Glue** for efficient **data processing**.
- Implemented **data quality frameworks** using **Apache Airflow and Great Expectations**, ensuring the **accuracy, consistency, and reliability** of data for **analytics and reporting**.
- Designed and deployed high-performance **data processing workflows**, identifying and resolving performance bottlenecks in distributed data systems to enhance processing speed and system reliability.
- Developed scalable **data architectures**, leveraging advanced **data modeling techniques** (conceptual, logical, and physical models) to optimize **storage, retrieval, and transformation efficiency**.
- Collaborated with **data scientists** and **business stakeholders** to deliver **high-impact data solutions**, ensuring seamless access to **structured and unstructured data** for **AI/ML-driven analytics**.
- Championed **best practices in data engineering**, enforcing robust **code quality, performance tuning, and security compliance** across all development cycles.
- Led **end-to-end development** of **big data infrastructure**, incorporating cutting-edge technologies and methodologies to enhance data availability, fault tolerance, and system resilience.
- Mentored and led **junior data engineers**, conducting code reviews, knowledge-sharing sessions, and architectural discussions to drive technical excellence and innovation.
- Automated large-scale **data ingestion pipelines**, ensuring **real-time and near-real-time data availability** for **mission-critical applications** using **Kafka, Spark Streaming, and AWS Kinesis**.
- Enhanced **cloud-based data infrastructure resilience**, ensuring **high availability, disaster recovery, and real-time failover mechanisms** for **uninterrupted data operations**.
- Optimized **data storage and retrieval strategies**, leveraging **columnar storage (Parquet, ORC), indexing, partitioning, and caching** to improve **query performance** and **reduce latency**.
- Designed and implemented **real-time reporting and analytics platforms**, integrating **BI tools, Apache Superset, and Elasticsearch** to provide **actionable business intelligence** for **decision-makers**.

TruWest Credit Union Senior Data Engineer

Jan 2018 – Jun 2023
Olympia, WA

- Worked with **G42 Cloud Elastic Cloud Service, Elastic Load Balancer, Cloud Container Engine, and Data Governance services** (including Data Arts Factory, Data Quality, Cloud Data Migration Services, Data Lake Insights, and Data Lake Visualization). Managed **Automated Jobs Configuration, MRS, Virtual Private Cloud, Relational & NoSQL Databases, Function Graph, and Cloud Backup & Recovery Services**.
- Experienced with **Hadoop distributions** from **Cloudera, Hortonworks, AWS, GCP, Azure ADF, G42 Cloud, Databricks, and Snowflake**—including administration tasks to ensure system performance.
- Designed, developed, and maintained **batch and real-time data pipelines** to automate **ETL (Extract, Transform, Load)** processes, enabling seamless data extraction from various sources into centralized repositories.
- Developed **real-time data ingestion** pipelines for **financial transactions** using **Apache Kafka** and **Apache Flink**, enabling teams to perform live monitoring, analysis, and reporting.
- Implemented comprehensive data quality checks and monitoring systems to ensure the **accuracy, completeness, consistency, and integrity** of data over time, driving confidence in data for decision-making.
- Identified and addressed **performance bottlenecks** in **data processing and storage systems**, achieving **optimal processing speed** and system **reliability**.
- Demonstrated a solid understanding of **data modeling** techniques (conceptual, logical, and physical models), designing efficient and scalable data models to optimize data flow.

- Worked closely with **data scientists, analysts, and business stakeholders** to understand their data needs, ensuring data is accessible, usable, and accurate for analysis and business insights.
- Spearheaded the **redesign of legacy data systems** into modern **data lakes** on **AWS**, utilizing **Apache Hive, Apache Spark, and S3** to enhance data accessibility and reporting efficiency.
- Reduced **pipeline processing time by 25%** through **optimization techniques** and **automation of data workflows** using **Apache Airflow**.
- Re-engineered legacy databases into a **Hadoop-based Data Lake**, leveraging **Spark, Hive, and Apache Nifi** for seamless data flow automation, resulting in a **40% improvement in data processing efficiency**.
- Collaborated with stakeholders to create **analytics applications** offering **real-time insights** for business decision-making.
- Managed **GitHub, GitLab, and SVN** repositories for streamlined version control across multiple projects.

US Bank
Data Engineer

May 2015 – Dec 2017
Minneapolis, MN

- Developed and managed **ETL pipelines** for diverse data sources, including **SQL, Oracle, CSV, JSON, and Kafka**, utilizing **Spark and Scala** to ensure efficient data processing and transformation.
- Designed, developed, and implemented multiple **Fraud Monitoring Systems** using **Apache Kafka, Cassandra, and Spark**, providing real-time alerts for proactive fraud detection.
- Worked extensively with **Google Cloud Platform** services such as **BigQuery, Google Data Studio, and Flow** to enable data analytics, reporting, and seamless integration of cloud-based data solutions.
- Wrote and maintained **complex ETL applications** utilizing **SSIS, SSRS, and SSAS cubes** to support diverse analytics and reporting needs across multiple departments.
- Collaborated closely with cross-functional teams to enhance **fraud detection and prevention controls** within telecom operations, driving the implementation of advanced methodologies for fraud identification.
- Conducted thorough **fraud risk assessments** for newly launched telecom products and services, playing a key role in **UAT evaluations** and defining **fraud prevention strategies**.
- Led the **management of data warehouse services** and the administration of big data clusters, leveraging tools such as **Apache Nifi, HDFS, Hive, Docker, and Postgres SQL**.
- Developed advanced **fraud detection rules** and **threshold-based alert systems**, improving the organization's ability to identify and mitigate fraud risks.
- Spearheaded the creation of **data visualization dashboards** using **Tableau and Kibana**, delivering actionable insights and supporting business decisions.
- Worked with multiple **open-source technologies** like **Apache Kafka, Nifi, Spark, and Cassandra** to ensure seamless data flow and processing across systems.
- Implemented **data governance services** and best practices to ensure **data integrity, security, and compliance** across various projects and platforms.
- Established 16 new **fraud control procedures** at **Warid Telecom**, addressing key fraud issues such as **CLIR, International Roaming Fraud, Duplicate Sims, Balance Share Fraud, and others**.
- Analyzed fraud patterns from **subscriber operations, calling trends, and sales channel activities**, leading to the creation of effective fraud prevention policies and strategies.
- Provided consultancy and technical support to **Sitronics Telecom Solutions**, helping improve the **RAFM tool features** and guiding the development of new solutions that added significant value.
- Coordinated with developers to provide logical frameworks for **new developments** in fraud management tools and supported **system support for fraud management** and reporting.

Certifications and Trainings

- AWS Certified Solutions Architect – Associate
- Big Data Engineering with Apache Spark (Udemy)
- Machine Learning Foundations (Coursera)
- Tableau (Udemy)

Education

Master's Degree in Computer Science

Rowan University | Jun 2013 – Apr 2015

Bachelor's Degree in Computer Science

Rowan University | Sep 2009 – May 2013